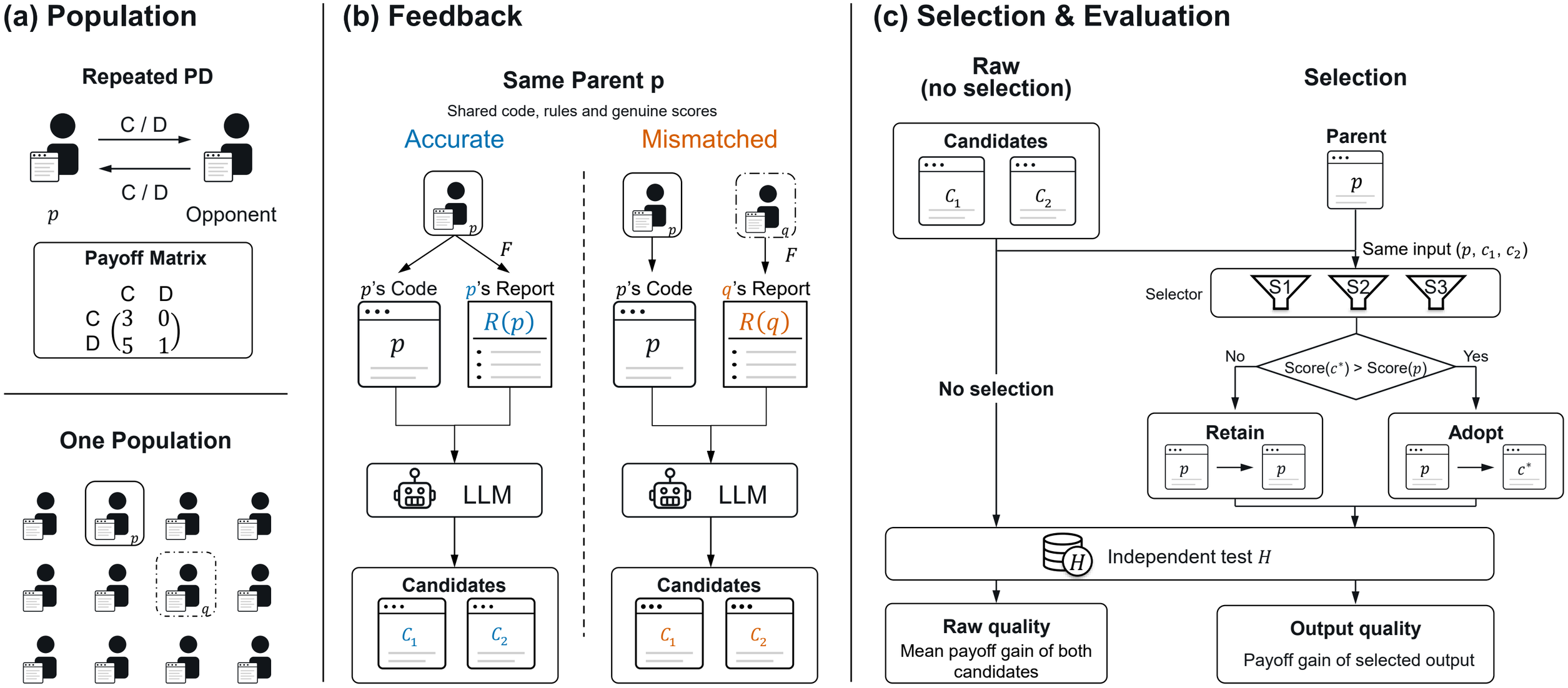

(a) Population
Repeated PD
C / D
C / D
p
Opponent
Payoff Matrix
C D
C (3 0)
D (5 1)
One Population
p
q
(b) Feedback
Same Parent p
Shared code, rules and genuine scores
Accurate
Mismatched
p
F
p's Code
p's Report
R(p)
LLM
Candidates
C1
C2
q
q's Report
R(q)
(c) Selection & Evaluation
Raw
(no selection)
Selection
Candidates
Parent
Same input (p, c1, c2)
Selector
S1
S2
S3
No
Score(c*) > Score(p)
Yes
No selection
Retain
Adopt
c*
Independent test H
Raw quality
Mean payoff gain of both candidates
Output quality
Payoff gain of selected output